HTML5 Exercises

1. Create a Page Using the Following Tags

Create an HTML web page that meets the following specifications.

Basic Structure

Create the basic structure of an HTML document using the following tags:

- `<!DOCTYPE html>`
- `<html lang="en">`
- `<head>` containing:
    - `<meta charset="UTF-8">`
    - `<meta name="viewport" content="width=device-width, initial-scale=1.0">`
    - `<title>My Web Page</title>`
- `<body>` containing:

Header

`<header>` with:

- `<h1>Welcome to My Web Page</h1>`
- `<p>This is a sample page for the exercise.</p>`

Sections

Add two sections to your page:

Main Section (<main>)

Include:

- `<h2>About Us</h2>`
- `<p>We are a company dedicated to providing web solutions.</p>`

Services Section (<section>)

Include:

- `<h2>Our Services</h2>`
- An ordered list `(<ol>)` containing three list items `(<li>)` describing the services offered by your company. The list should use alphabetical numbering.
- A hyperlink `(<a>)` with the text **"View More Services"** that points to:

https://www.example.com

The link must open in a new browser tab or window.

Footer

Create a footer `(<footer>)` containing:

- A hyperlink `(<a>)` with the text **"Send Email"** and the attribute:

`href="mailto:info@mycompany.com"`

- A hyperlink `(<a>)` with the text **"Call by Phone"** and the attribute:

`href="tel:+1234567890"`

- Include the special character **©** followed by the name of your company and the text **"All Rights Reserved"**. Use an HTML entity, hexadecimal code, or decimal code to insert the copyright symbol.

2. Create a Registration Form

Create an HTML registration page that collects detailed information from users. The page must contain the following fields:

Personal Information

- First Name (text field, required)
- Last Name (text field, required)
- Date of Birth (date field, required)
- Gender (radio buttons: Male, Female, Other; required)

Contact Information

- Email Address (email field, required)
- Phone Number (telephone field, required)
- Address (multi-line text area)

Account Information

- Username (text field, required)
- Password (password field, required)
- Confirm Password (password field, required)

Preferences

- Favorite Color (drop-down/select list: Red, Blue, Green, Yellow)
- Accept Terms and Conditions (checkbox, required)

Use the `<fieldset>` and `<legend>` tags to group and describe sections of the form.

Ensure that all fields marked as **required** must be completed before the form can be submitted.

Link each `<label>` element to its corresponding input field using the for and id attributes to improve accessibility.

3. Create a Table of Contents in HTML5

Imagine you are creating a web page for an online book and need to create a table of contents to display the book's chapters.

Use the following HTML tags as needed:

`<table>, <caption>, <tr>, <td>, <th>, <colgroup>, <col>, <tbody>, <thead>, <tfoot>`

and the attributes: `rowspan` and `colspan`.

Instructions

- Create a `<table>` that will contain the book's table of contents.
- Add a table header section `(<thead>)` containing a row `(<tr>)`.
- In the header row, add header cells `(<th>)` for the following columns:
    - Chapter
    - Chapter Title
    - Pages
- Use the colspan attribute to merge the three header cells into a single header cell containing the text:

"Book Contents"

- Add three rows `(<tr>)` inside the table body `(<tbody>)` to represent the first three chapters of the book.
- Inside each row, add data cells `(<td>)` for:
    - Chapter
    - Chapter Title
    - Pages
- Fill the cells with sample information for the first three chapters.
- Use the rowspan attribute to merge chapter cells where appropriate. This should indicate how many rows a chapter occupies.
- Add an additional row `(<tr>)` to represent the fourth chapter. This chapter should occupy only one row.
- Complete the table by adding more rows and data cells to represent additional chapters while maintaining the same table of contents structure.

4. Create a Web Page Using Embedded Content and Bookmarks

Create a web page that includes the following features:

Display an Image

Use the `<img>` tag to display an image on your page. Include the following attributes:

- src: Specifies the URL or file path of the image.
- alt: Provides alternative text if the image cannot be displayed.
- width: Sets the image width in pixels.
- height: Sets the image height in pixels.

Embed a YouTube Video

Insert a YouTube video into your page using the <iframe> tag.

Follow these steps:

1. Open the YouTube video you want to embed.
2. Click the **Share** button below the video.
3. Select **Embed**.
4. Copy the HTML code provided by YouTube and paste it into your web page.

Add a Page Title

Use the `<h1>` tag to add a title or heading to your page. Choose a title relevant to your content.

Create Internal Navigation Links

Use bookmarks to create internal links that allow users to quickly navigate between the image and the video.

Use the <a> tag with the href attribute pointing to the appropriate bookmark locations.

Add Metadata

Include metadata tags such as:

```
<meta name="description" content="Page description">
```

```
<meta name="author" content="Your Name">
```

Additional Content

You may add any additional text or content you wish to provide context and improve the overall presentation of your page.